

QUESTION PAPER

1. How do metaheuristics help solve complex optimization problems?
1. What role does social interaction play in Particle Swarm Optimization and its convergence to an optimal solution?
2. Explain the role of pheromone evaporation in preventing stagnation in Ant Colony Optimization(ACO).
2. How Ant Colony Optimization (ACO)can be adapted for dynamic optimization problems?
3. Write the Principles of Bird Flocking.
3. What challenges are typically faced when applying PSO to high-dimensional optimization problems?
4. How does the ABC algorithm utilize food sources to represent solutions in an optimization problem?
4. Differentiate employed bees and scout bees in the ABC algorithm and their respective roles in the search process.
5. How is the fitness of a solution evaluated in the Krill Herd Optimization process?
5. How does Krill Herd Optimization help in improving the accuracy and efficiency of feature selection in machine learning?

PART B

6. Demonstrate how the Krill Herd Algorithm can be adapted to solve a real-world optimization problem. What modifications would you make to improve its performance?
OR
i).Discuss the advantages and limitations of Best First Search. ii).Explain in detail about A* and Hill Climbing .
7. Evaluate the strengths and limitations of ACO. Discuss possible enhancements or hybridizations to improve the performance of ACO in complex optimization problems.
OR
Elaborate on the role and significance of pheromone evaporation and reinforcement mechanisms in preventing premature convergence in ACO variations.
8. Discuss the evolution of Particle Swarm Optimization (PSO). How did the algorithm evolve from its initial concept to the modern variations?
OR
i).Describe the concept of 'fitness function' in PSO. ii). How is the fitness of a particle evaluated, and how does this evaluation guide the search for an optimal solution?
9. Compare the basic ABC algorithm with its variations, ABCgbest and ABCgbestdist. How do these variations affect the exploration and exploitation balance in the optimization process? Discuss the advantages and disadvantages of using ABCgbestdist over the standard ABC.
OR
Explain how the Artificial Bee Colony (ABC) algorithm can be extended to handle multi-objective optimization problems.
10. How can you adjust the parameters of the Krill Herd Optimization algorithm to enhance its convergence speed without compromising solution quality?
OR
Apply the Krill Herd Algorithm to a multi-objective optimization problem and describe the modifications required for handling multiple objectives effectively.

PART C

11. Evaluate how Ant Colony Optimization (ACO) can be integrated with other optimization algorithms
OR

Explain the challenges of using Swarm Intelligence for Knapsack Problem and Feature Selection, such as local optima, parameter tuning, and high-dimensional search spaces.